A variance bound for Davis's crowd-estimation polynomials

Research proof and Lean formalization
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Abstract

We prove the rational-function bound proposed in Conjecture 3.1 of Davis's paper on crowd estimation. The inequality holds strictly throughout its stated domain. The proof rewrites the polynomial quotient as a variance, derives an exact fourth-moment identity by finite detailed balance, and uses a quadratic comparison for every integer degree at least 43. The remaining 38 degrees are covered by exact positive-coefficient certificates. The source's separate conjecture on critical points is not addressed.

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1 Statement and the finite distribution

For $i \geq 0$ write $f_i(x) = \prod_{j=0}^{i-1}(x-j)$, with $f_0 = 1$. Define

$$P_d(x) = \sum_{i=0}^{d} \frac{f_i(x)^2 f_{d-i}(x)^2}{i!(d-i)!}, \tag{1}$$

$$q_d(x) = \frac{2x^2 - 2dx}{d+1} + \frac{d}{2}, \qquad R_d(x) = P_{d+1}(x) - q_d(x)P_d(x). \tag{2}$$

Davis proves that q_d is the polynomial quotient and R_d the remainder (Lemma 1.6 and equation (2.4) in the arXiv version; Lemma 1.1 and equation (2.4) in the revised journal-style manuscript) [1]. His Conjecture 3.1 is the following inequality with a weak conclusion.

Theorem 1. *For every integer $d \geq 5$ and real $x \geq (d + \sqrt{d+2})/2$,*

$$-\frac{R_d(x)}{P_d(x)} > \frac{199}{200}\,\frac{d-2}{2x+d-2}.$$

The constant is the source's 0.995. We make no claim that it is optimal.

Put $w_i = f_i(x)^2 f_{d-i}(x)^2/(i!(d-i)!)$ and $Z = P_d(x)$. These weights are nonnegative and $Z > 0$ when $x > d/2$: the term $i = \lfloor d/2 \rfloor$ has only positive factors in both falling factorials. Let expectations use probabilities w_i/Z, and set

$$D = d/2, \quad a = x - D, \quad u = i - D, \quad V = \mathbb{E}u^2, \quad M = \mathbb{E}u^4.$$

Symmetry gives $\mathbb{E}i = D$. If w'_j denotes the degree-$(d+1)$ weights, then $(i+1)w'_{i+1} = (x-i)^2 w_i$. Summing and using symmetry again yields

$$\frac{P_{d+1}}{P_d} = \frac{2}{d+1}\mathbb{E}(x-i)^2 = \frac{2}{d+1}(a^2+V), \qquad -\frac{R_d}{P_d} = \frac{d-4V}{2(d+1)}. \tag{3}$$

Thus an upper bound for V suffices.

2 An exact fourth-moment identity

For $0 \leq i \leq d$ put

$$\lambda_i = (d-i)(x-i)^2, \qquad \mu_i = i(x-d+i)^2.$$

Direct multiplication gives $\lambda_i w_i = \mu_{i+1} w_{i+1}$ for $i < d$, and $\lambda_d = \mu_0 = 0$. Hence, for any function h on the integers,

$$\sum_{i=0}^{d} w_i\{\lambda_i[h(i+1) - h(i)] + \mu_i[h(i-1) - h(i)]\} = 0. \tag{4}$$

This is a finite index-shift identity. No division by $x - i$ is used, so vanishing weights at integer inputs cause no exception.

Take $h(i) = (i - D)^2$. Expansion gives

$$\lambda - \mu = -2u(2Da + a^2 + u^2), \qquad \lambda + \mu = 2(Da^2 + Du^2 + 2au^2).$$

Substituting into (4) and dividing by $2Z$ proves

$$2M + BV = Da^2, \qquad B = 4Da - D + 2a^2 - 2a. \tag{5}$$

Since $\mathbb{E}(u^2 - V)^2 \geq 0$, we have $M \geq V^2$. Therefore

$$2V^2 + BV \leq Da^2. \tag{6}$$

3 Degrees at least 43

The source domain implies $a \geq \sqrt{d+2}/2 > 1$ and $4a^2 \geq d + 2$. Define

$$T = \frac{d}{4} - \frac{199(d+1)(d-2)}{800(a+d-1)} = \frac{200da + d^2 - d + 398}{800(a+d-1)} > 0.$$

By (3), the conclusion of Theorem 1 is precisely $V < T$. Also $B > 0$. The needed comparison is certified by

$$\begin{aligned} 2T^2 + BT - Da^2 &= \frac{N(d,a)}{320000(a+d-1)^2}, \\ N(d,a) &= 800K(a+2d-2)\left(a^2 - \frac{d+2}{4}\right) + aL(d) + K(201d^2 + 599d - 402), \\ K &= d^2 - d + 398, \\ L(d) &= 400(2d^4 - 105d^3 + 902d^2 - 1197d + 1194). \end{aligned}$$

Every term in N is nonnegative, and the last two are positive. Indeed $K > 0$, and for $t = d - 43 \geq 0$,

$$L(d) = 800t^4 + 95600t^3 + 3818000t^2 + 51998400t + 42755200 > 0.$$

Thus $2T^2 + BT > Da^2$. On $[0, \infty)$ the polynomial $2v^2 + Bv$ is strictly increasing. Since $V \geq 0$, inequality (6) forces $V < T$. This proves the entire infinite range.

4 Degrees 5 through 42

Let $S_d = d!P_d$, an integer polynomial. To avoid fractions define

$$\begin{aligned} H_d(x) = \{&200(2x + d - 2)[4x^2 - 4dx + d(d+1)] - 398(d+1)(d-2)\}S_d(x) \\ &- 400(2x + d - 2)S_{d+1}(x). \end{aligned}$$

An exact identity is

$$H_d = 400(d+1)d!\left\{[(2x + d - 2)q_d - (199/200)(d-2)]P_d - (2x + d - 2)P_{d+1}\right\}. \tag{7}$$

Since $P_d > 0$ and $2x + d - 2 > 0$, positivity of H_d is equivalent to the strict desired bound.

Put

$$m_d = 1000d + \left\lfloor \sqrt{10^6(d+2)} \right\rfloor, \qquad r_d = m_d/2000 \leq (d + \sqrt{d+2})/2.$$

Represent H_d by its ascending integer coefficient list A_d; trailing zeros are permitted. If its list length is ℓ_d, set

$$B_d(y) = 2000^{\ell_d - 1} H_d(y + r_d).$$

For each of the 38 integers $d = 5, \ldots, 42$, the constant coefficient of B_d is positive and every coefficient is nonnegative. These are exact integer certificates, checked from the defining falling products. Consequently $B_d(y) > 0$ for $y \geq 0$, hence $H_d(x) > 0$ for every $x \geq (d + \sqrt{d+2})/2$. Together with the preceding section this proves Theorem 1.

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5 Formal certificate and scope

The Lean development uses the literal finite sum (1) and difference (2). It proves positivity of P_d, reflection and the mean identity, detailed balance, the fourth-moment equation, $M \geq V^2$, and the uniform scalar comparison. There are no residual moment or polynomial-identity hypotheses in the source conclusion. The endpoint names are

`Davis.source_strict_bound`, `Davis.source_conjecture_319187`.

The finite checker constructs all coefficients internally from the source formulas. Polynomial addition, multiplication, scalar multiplication, and the affine translation are implemented on integer lists, with their evaluation identities proved in Lean. Binomial coefficients use the factorial implementation whose equality with the usual binomial coefficient is proved in mathlib. The 38 positivity checks use kernel reduction of integer arithmetic. They do not invoke an untrusted external numerical oracle. The rational shift is bounded using the proved inequality $\lfloor\sqrt{n}\rfloor^2 \leq n$.

The certificate has 100 declarations in 19 modules. A fresh isolated replay with Lean 4.34.1 and mathlib revision `d13f23b7` passes with warnings as errors. Every declaration type and axiom report was inspected; only `propext`, `Classical.choice` and `Quot.sound` occur.

As supplementary checks, a pure-integer Python implementation and a separate symbolic rational-polynomial implementation produce positive certificates for all 38 finite degrees. Exact rational evaluation verifies the weight, quotient and moment identities at 174 inputs, including 45 cases with zero weights. These checks are independent of the universal mathematical argument and are not used to replace the Lean proof.

The source's established half-unit bound, polynomial quotient, conditional crowd-estimation consequences, and numerical observations are credited to Davis. Detailed balance and the variance inequality are standard tools; the proposed contribution is their combination with the explicit uniform comparison and finite certificates to resolve the exact source conjecture. We do not prove the distinct critical-point Conjecture 3.9 of the arXiv version (Conjecture 3.2 in the revised manuscript), identify the optimal constant, or claim new general moment theory.

The prior-art investigation is bounded. The current arXiv entry has only version1, the author's revised 2010 manuscript retains Conjecture 3.1, and targeted searches located no resolution. The exact OpenAlex record has no indexed citers; this count is not evidence that no unindexed or unpublished resolution exists. Subscription indexes and a final paginated publisher copy were not available. No absolute priority claim is made.

References

[1] D. M. Davis, *Maximizing a combinatorial expression arising from crowd estimation*, International Journal of Modern Mathematics **5** (2010),129–148. arXiv:0910.5725v1. Author-hosted revised journal-style manuscript (May 14, 2010): https://www.lehigh.edu/~dmd1/fredijmm.pdf.

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